

Numerical Approach: Newton-Raphson

$$D' = 2(x - x_0) + 2(x^2 + 5 - y_0)(2x)$$

$$\begin{aligned} D'' &= 2 + 2[(2x)(2x) + (x^2 + 5 - y_0)(2)] \\ &= 2 + 6x^2 + 4x^2 + 20 - 4y_0 \\ &= 8x^2 - 4y_0 + 22 \end{aligned}$$

At $x_0, y_0 = (0, 0)$:

$$D' = 2x + 2(x^2 + 5)(2x) = 2x + 4x(x^2 + 5)$$

$$D'' = 8x^2 + 22$$

start with guess $x = 0$:

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = 0 - \frac{2(0) + 4(0)(0^2 + 5)}{8(0)^2 + 22}$$

$$0 = 0 - 0 \quad \checkmark \quad \text{Found exact answer.}$$

Iteration 1

$$x = 0$$

$$y = (0)^2 + 5 = 5$$

$$d = \sqrt{0^2 + 5^2} = 5$$

At $x_0, y_0 = (-4, 0)$:

$$D' = 2(x + 4) + 4x(x^2 + 5)$$

$$D'' = 8x^2 + 22$$

start with guess $x = 0$:

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = 0 - \frac{2(0 + 4) + 4(0)(0^2 + 5)}{8(0)^2 + 22}$$

$$x^* = 0 - \frac{8}{22} \quad (-0.36)$$

$$x^* = -\frac{8}{22} \quad \times \quad \text{Try } x = -8/22.$$

Iteration 1

$$D' = 2(x + 4) + 4x(x^2 + 5)$$

$$D'' = 8x^2 + 22$$

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = \frac{-8}{22} - \frac{2(-\frac{8}{22} + 4) + 4(-\frac{8}{22})((-\frac{8}{22})^2 + 5)}{8(-\frac{8}{22})^2 + 22}$$

$$x = -0.355 \rightarrow \text{same as what was solved via analytical}$$

Iteration 2

stop here.

$$x = -0.355$$

$$y = (-0.355)^2 + 5 = 5.126$$

$$d = \sqrt{(-0.355)^2 + 5.126^2} = 5.138$$

At $x_0, y_0 = (-8, 0)$:

$$D' = 2(x+8) + 4x(x^2+5)$$

$$D'' = 8x^2 + 22$$

Start with guess $x = 0$.

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = 0 - \frac{2(0+8) + 4(0)(0^2+5)}{8(0)^2 + 22}$$

Iteration 1

$$x^* = 0 - \frac{16}{22}$$

$$x^* = -\frac{8}{11} \quad (-0.727) \quad \text{Try } x = -8/11$$

$$D' = 2(x+8) + 4x(x^2+5)$$

$$D'' = 8x^2 + 22$$

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = -\frac{8}{11} - \frac{2(-\frac{8}{11} + 8) + 4(-\frac{8}{11})((-\frac{8}{11})^2 + 5)}{8(-\frac{8}{11})^2 + 22}$$

$$x = 0.669$$

Iteration 2:

Stop here.

$$x = 0.669$$

$$y = (0.669)^2 + 5 = 5.448$$

$$d = \sqrt{0.669^2 + 5.448^2} = 5.448$$

At $x_0, y_0 = (2, 0)$:

$$D' = 2(x-2) + 4x(x^2+5)$$

$$D'' = 8x^2 + 22$$

Start with guess $x = 0$.

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

Iteration 1

$$x^* = 0 - \frac{0 - 2(0-2) + 4(0)(0^2+5)}{8(0)^2 + 22}$$

$$x^* = 0 - \frac{2}{11}$$

$$x^* = -2/11 \quad \times \quad \text{Try } x = -2/11 \quad (0.18)$$

$$x^* = -2/11 - \frac{2(-2/11-2) + 4(-2/11)((-2/11)^2+5)}{8(-2/11)^2 + 22}$$

$$= 0.179$$

Iteration 2: stop here.

$$x = 0.179$$

$$y = 0.179^2 + 5 = 5.032$$

$$d = \sqrt{0.179^2 + 5.032^2} = 5.035$$

At $x_0, y_0 = (6, 0)$:

$$D' = 2(x-6) + 4x(x^2+5)$$

$$D'' = 8x^2 + 22$$

Start with guess $x = 0$.

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = 0 - \frac{2(0-6) + 4(0)(0^2+5)}{8(0)^2 + 22}$$

$$x^* = 0.545$$

Iteration 1

$$x^* = 0.545 \quad \times \quad \text{Try } x = 0.545$$

$$D' = 2(x-6) + 4x(x^2+5)$$

$$D'' = 8x^2 + 22$$

$$x^* = x_0 - \frac{f'(x_0)}{f''(x_0)}$$

$$x^* = 0.545 - \frac{2(0.545-6) + 4(0.545)(0.545^2+5)}{8(0.545)^2 + 22}$$

$$x^* = 0.519$$

Iteration 2:

Stop here.

$$x = 0.519$$

$$y = (0.519)^2 + 5 = 5.269$$

$$d = \sqrt{0.519^2 + 5.269^2} = 5.294$$